
kmbox Net User Manual

What is Kmbox Net?

kmbox Net is a network-controlled keyboard and mouse controller with the following features:

- * 1. Secure, protocol not publicly disclosed. Each box has an independent IP, port, and hardware code. Utilizes blocking socket communication, making it unscannable.
- * 2. High stability, unlike serial communication which can cause white screen and restarts. * 3. High speed. A dedicated 100M network provides 100 times the communication speed of the original B board (compared to 115200 baud rate). It calls nearly 1000 times per second, compared to only 300 times per second on the B board, which experiences white screen and restarts due to faster calls. * 4. Automatically defaults to manual tracking, preventing abnormal keyboard and mouse data. *
- 5. No mouse adapter required, highly versatile. * 6. Device-side supports modification of all USB parameters, ensuring unique characteristics for each user, truly a one-person-one-firmware model. * 7. User-friendly UI mode, problems are clearly identified, and operation is intuitive. * 8. Supports all functions of the B board except for onboard scripts. * 9. Supports monitoring of physical keyboard and mouse, and disabling physical keyboard and mouse functions for easy software development. * 10. Supports physical keyboard and mouse learning and copying functions. Supports driver passthrough mode.

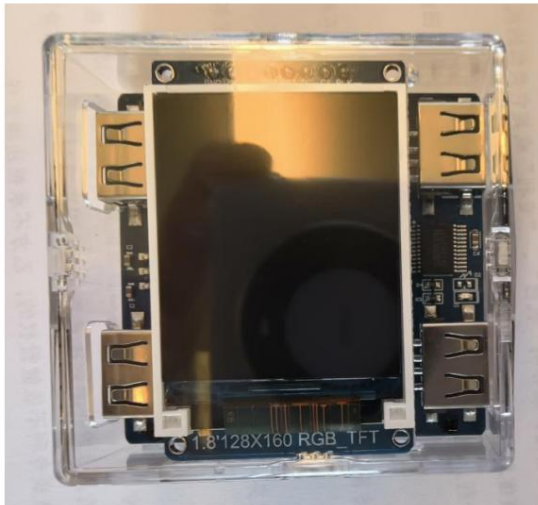
Beginner's Guide [First-time users please see here]

This section will provide a detailed explanation of how to use Kmbox Net.

01 Unboxing Introduction

kmboxNet includes the following accessories by default.

- kmbox Net box



* Two blue cables



02 How to connect the wiring box: The interface names are on the back of the box. Generally, connect the wiring according to the following diagram:



Note : If it's a standalone game, connect both the network port and the game console port to the same computer. If computer B controls computer A, then connect game console port (1) to computer A and network port (2) to computer B.

03 Install network card driver (install driver for computers with 2 ports)

After connecting the network cable, a disk will appear in My Computer, as shown in the image below. Double-click the disk to find the .exe file and install it. If there is no disk, you can manually download the network card driver, extract it, and install it. Click here to [download the network card driver](#).

Note: If installing the network card driver has no effect: (1) Check if the network cable is connected? (2) Install the network card driver where the network port USB cable is connected. Do not install the driver on computer A and connect the cable to computer B.



04 Change the network card IP (Change the network card IP of a computer with a 2-port connection)

Open Control Panel --> Network and Sharing Center

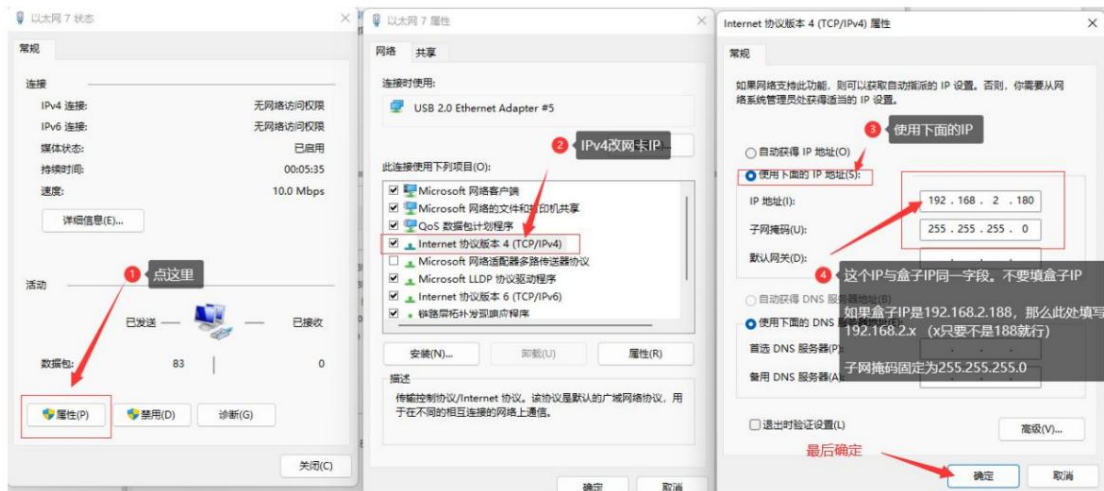




[Click here to download network card driver](#)

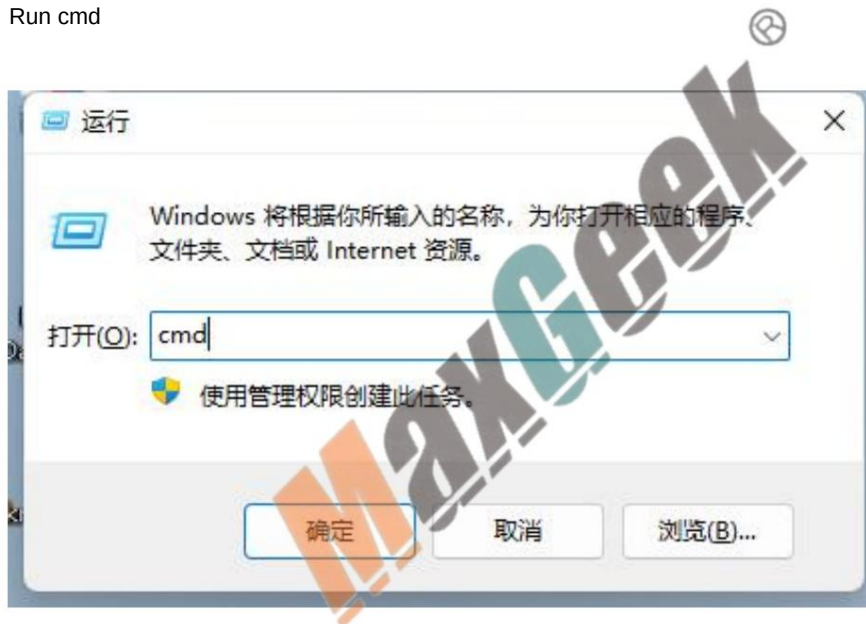
Change network card IP





Check if the network is connected.

Run cmd



Ping the IP address displayed on the box screen

ping 192.168.2.188



```
管理员: C:\Windows\system32\cmd.exe
Microsoft Windows [版本 10.0.22000.2416]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\Administrator>ping 192.168.2.188

正在 Ping 192.168.2.188 具有 32 字节的数据:
来自 192.168.2.188 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.2.188 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.2.188 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.2.188 的回复: 字节=32 时间<1ms TTL=128

192.168.2.188 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
    往返行程的估计时间(以毫秒为单位):
        最短 = 0ms, 最长 = 0ms, 平均 = 0ms

C:\Users\Administrator>
```

已ping通。说明此时电脑和盒子已经可以正常连接了

If the above image appears, it means the box is now set up. The next step is simply to configure third-party software to connect to the box.

The above process can be referenced in this video tutorial:

Note: If you suddenly lose internet access after changing the IP address, it's likely due to incorrect network adapter settings. A new network adapter (the box's network adapter) will usually appear on the page; you need to change the box's network adapter, not your own. See the network adapter modification tutorial above for detailed instructions.

Basic Tutorial [Introduction to the Box's Basic Functions]

Display Screen Specifications



As shown in the figure:

1. Game console port status icon

The game console status icon indicates the relationship between the box and the game console. This icon has three states.

Waiting for connection  Waiting for the console (game machine) to enumerate the box. This status indicates that the box has not yet been recognized by the console.

The host has successfully recognized the box and is now connected. The host can be controlled from the box. 

Disconnected status  The host computer disconnected from the box. Keyboard and mouse are unusable.

Note: The box is always a USB slave device. The box will never disconnect from the host. All connections and disconnections are initiated by the host.

2. Network port connection status icon

The network port status icon is used to display the connection relationship between the box and third-party software. This icon has three states.

The box is in a waiting connection state. At this time, the box is idle and no software is connecting to it.



Connected

A checkmark indicates that third-party software has been connected to the box.



A cross indicating a disconnection status means the third-party software is malfunctioning and has disconnected from the box.



This icon clearly indicates whether third-party software is connected to the box.

3. Keyboard and mouse port status

The keyboard and mouse button on the right side of the box is for connecting an external keyboard or mouse. There are a total of 8 states.

Waiting for mouse connection

This port is currently idle and awaiting external device connection. A mouse is recommended for this connection.



Mouse connected

A checkmark indicates that the peripheral device has been successfully recognized as a mouse and can be used.



Mouse disconnect

A cross indicates that the peripheral device is disconnected from the box. This is usually caused by an unstable connection.



Waiting for keyboard connection

This port is currently idle and awaiting external device connection. A keyboard is recommended for this connection.



Keyboard connected

A checkmark indicates that the peripheral device has been successfully recognized as a keyboard and can be used.



Keyboard disconnected

A cross indicates that the peripheral device is disconnected from the box. This is usually caused by an unstable connection.



Waiting for mouse
connection



This port is currently idle and awaiting external device connection. A mouse is recommended for this connection.

The combo keyboard and mouse
are connected.



The current peripheral device includes both keyboard and mouse reports, making it a combined keyboard and mouse device. And it has been successfully recognized.

Composite device
recognition failed



Unrecognized composite device. This is generally uncommon. If it does occur, please contact the author.

4. Inherent parameters of the box



This parameter is used by third-party software to connect to the box. Third-party software must enter the parameters corresponding to the box to connect; if even one is incorrect, the connection will fail.

IP: The IP address of the box.

Port: The service port number of the box.

UUID: A unique identifier for the box. Each device has a unique identifier.

5. Curve Type

The box has built-in curve types.



There are currently 5 values available, which can be switched using the host computer software.

Auto

Automatic curve inside the box

Bezier - Non-real-time Bezier curves

RT-Bezier - Real-time Bezier Curves

Arm - Non-real-time missile tracking curve

RT-Arm - Real-time tracking curves

The box has curves disabled by default, or only performs minor corrections. To enable curves, please refer to the curve modification tutorial.

6. Firmware version number



The firmware version shown in the image is November 6, 2025. The suffix letters are defined as follows:

Box firmware version naming rules:

Firmware release date + English suffix.

For example: 20251106A

The firmware release date is November 6, 2025.

A indicates automatic curve correction.

B indicates that the bypass version is supported.

C indicates a customer-customized version.

D indicates the Debug version.

R indicates the release version.

7. Box Model



8. Box Working Mode

The box currently has two working modes:

I. KM Composite Keyboard and Mouse Mode

The box operates in this mode by default.



In this mode, all keyboard and mouse functions are available. There are no restrictions. Peripherals connected to the box will be converted into reports by the box. These reports are then sent to the host device via the box. The box acts as an isolation device.

II. Bypass Driver Pass-through Mode

This mode can be enabled using host computer software.



In simple terms, the driver pass-through function means that the box can 100% transmit all data from the keyboard or mouse connected to it. There is no difference between connecting the box and connecting it to a computer.

Advantages of drive pass-through

1. It can use the original driver and supports the original driver macro.
2. It can 100% restore the characteristics of peripheral devices.

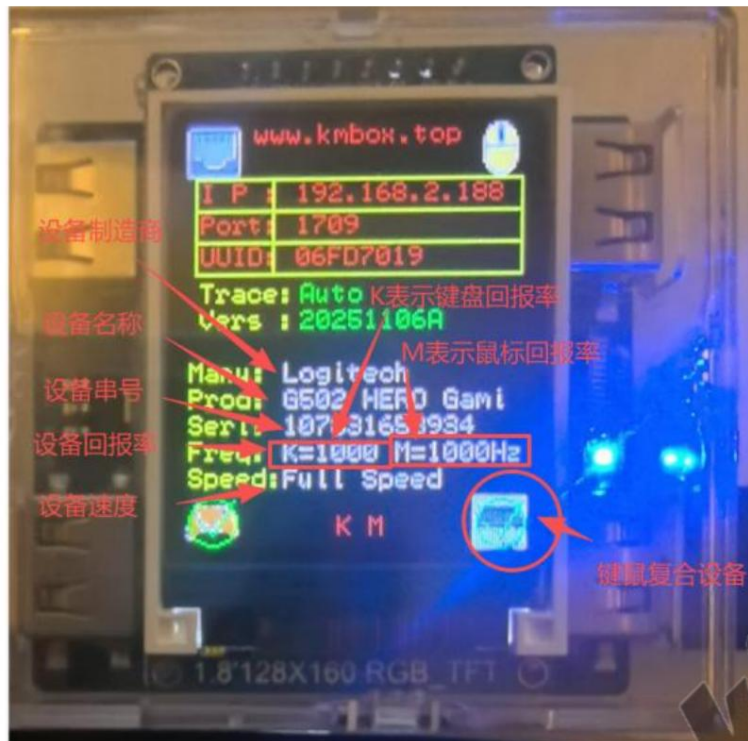
Disadvantages of driver pass-through

1. For keyboard-only devices, after driver passthrough, mouse movement commands cannot be invoked via API.
2. For pure mouse devices, after driver passthrough, it is impossible to call keyboard commands via API.

It's important to emphasize that current detection methods are all behavioral. Simply enabling driver passthrough doesn't bypass the detection. If your third-party software sends unreasonable commands, it will trigger behavioral anomalies. Driver passthrough merely replaces the device with one exhibiting the abnormal behavior. Those who understand, understand.

9. Peripheral Information

This section displays the basic parameter information of the latest device connected to the box. See the image below:



Focus on return on investment (Freq) and device speed (Speed).

In Freq, K=0 or M=0 may appear. This indicates that the external device is simply a keyboard or mouse.

If both K and M have values, it means your device has both keyboard and mouse reports.

Speed indicates your device's speed level. Currently, Net supports two speed levels.

1. Low Speed. 1.5Mbps bandwidth.

This indicates that your keyboard or mouse is a low-speed device. It's a relatively low-end keyboard and mouse. If you're playing games, I recommend you get a different mouse.

2. Full Speed. 12Mbps bandwidth.

This means your keyboard or mouse is a full-speed device. Most keyboards and mice are like this. Their speed is 10 times that of low-speed devices.

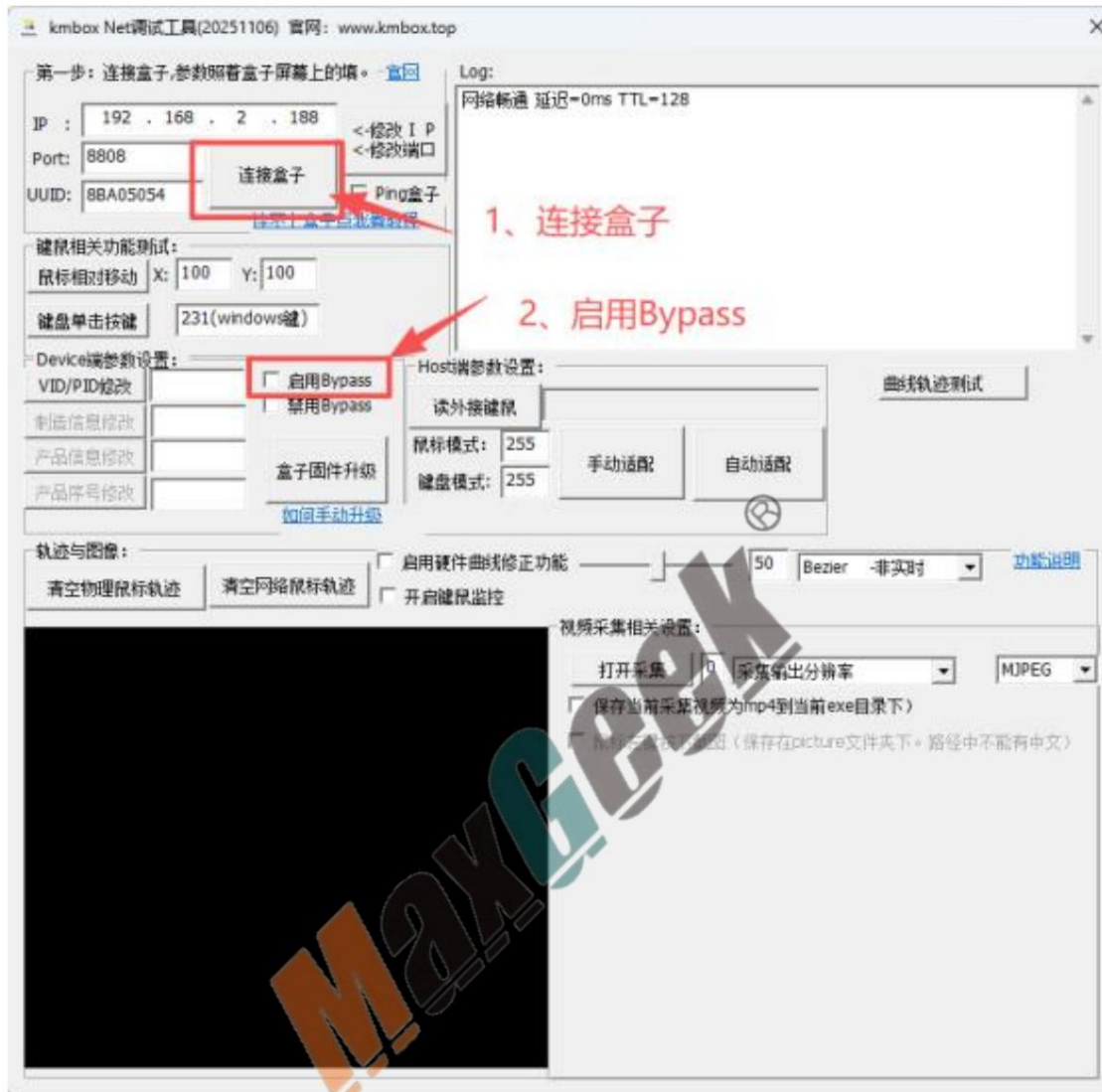
3. High Speed. 480Mbps bandwidth.

This is a high-end keyboard and mouse with a polling rate of up to 8kHz. Its response speed is 8 times faster than Full Speed. It is supported by the Kmbox notebook board.

Macro keyboards and mice typically include both keyboard and mouse reports.

How to enable bypass mode

Open the host computer software for the box and connect to the box first. Then click "Enable Bypass," and the box will automatically restart and enter driver pass-through mode.



Note: In drive pass-through mode, only one device can be connected to the box at a time.

If two devices are connected, the device that is directly connected to Port 4 will be driven first. Devices connected to Port 3 can be merged into devices connected to Port 4.

For example, a Logitech mouse connected to Port 4 can use the Ghub driver normally.

If you connect another keyboard or mouse to Port3, the keyboard and mouse data from Port3 will be converted into Logitech mouse protocol data. This is essentially a fusion function.

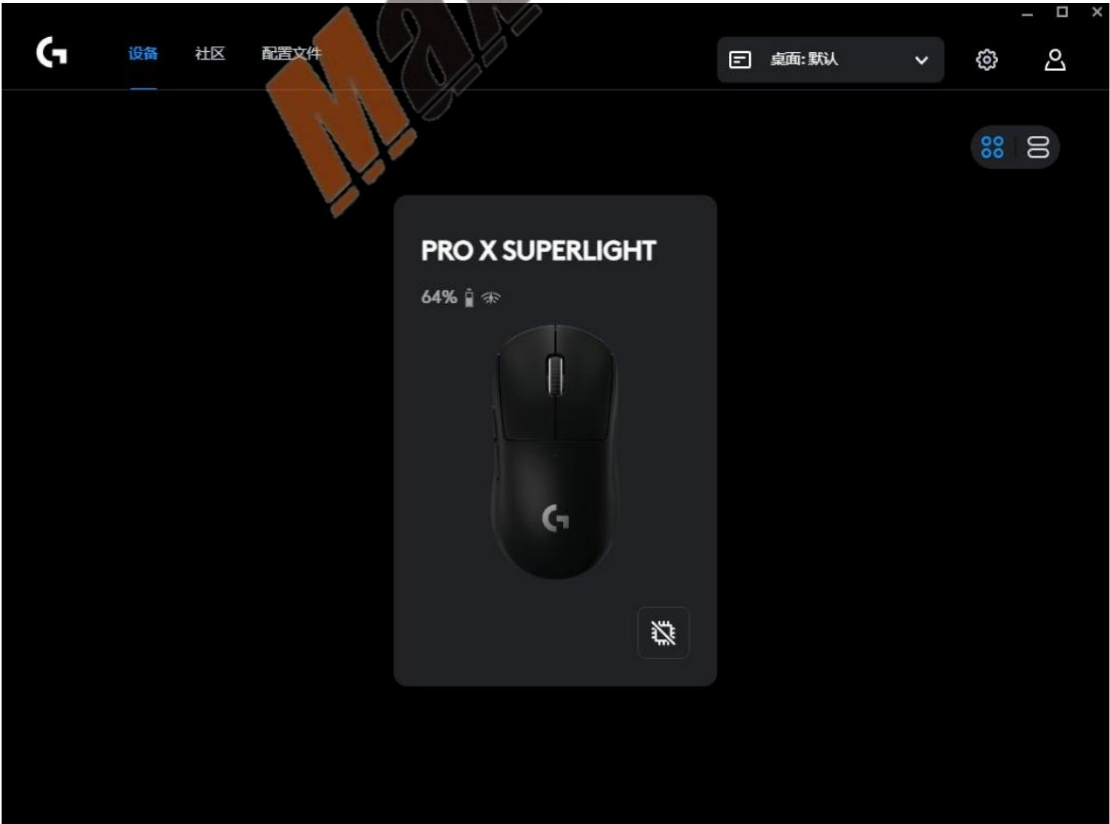
If the driver passthrough mode is not working properly, please provide feedback, and a firmware update will resolve the issue. You can disable Bypass mode and use KM mode normally.

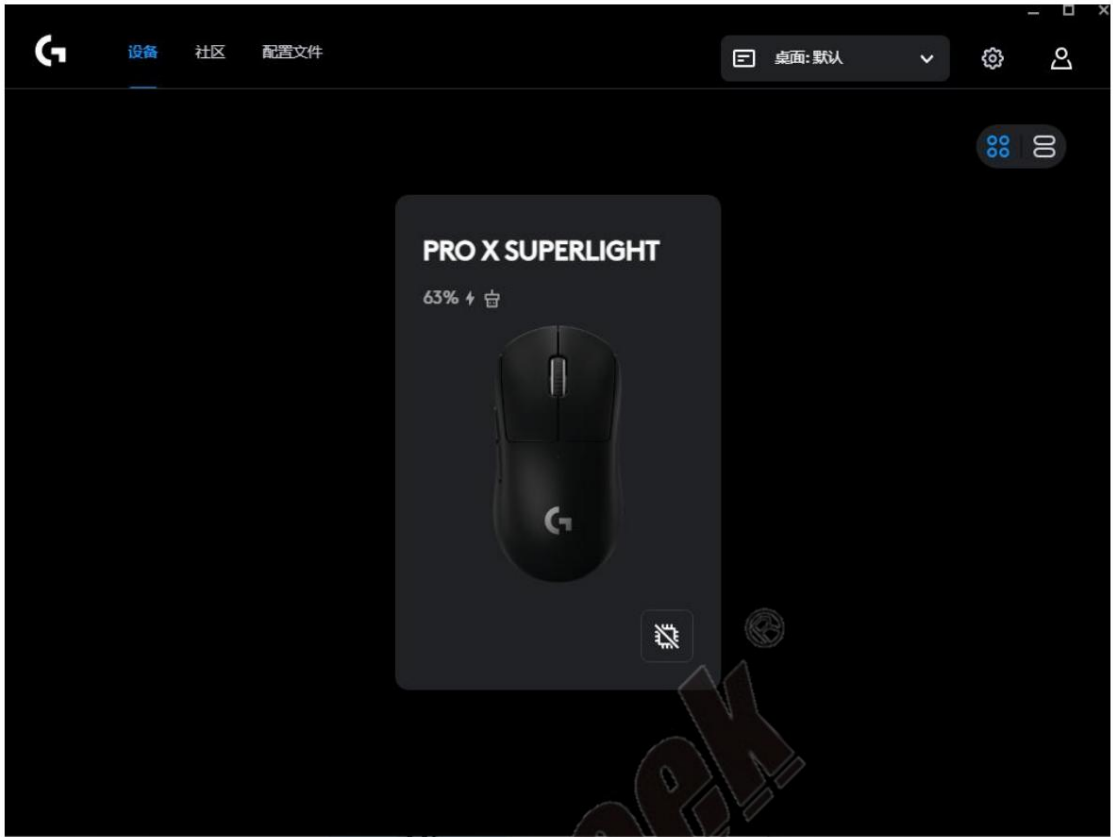
The driver passthrough feature currently supports the vast majority of mice. Please report any unsupported mice. We will update this feature in future upgrades. Below are screenshots of some of the tested devices:

Logitech series:



GPW series (wireless + wired both available)





Maicong series: (Both web-based and PC-based drivers are compatible)





Rapoo series (8kHz polling rate)



Theoretically, driver passthrough for any type of keyboard and mouse is already supported. For other untested types, please test them yourself. If you encounter any issues, please provide feedback.

How to enable KM mode

For users who need simultaneous keyboard and mouse input, the Bypass mode can be disabled. The device is a keyboard and mouse combo device by default, allowing access to both keyboard and mouse functions.



How to upgrade

kmbox Net updates its firmware irregularly every month. You can download the minimum firmware and upgrade tools from the official website to upgrade.

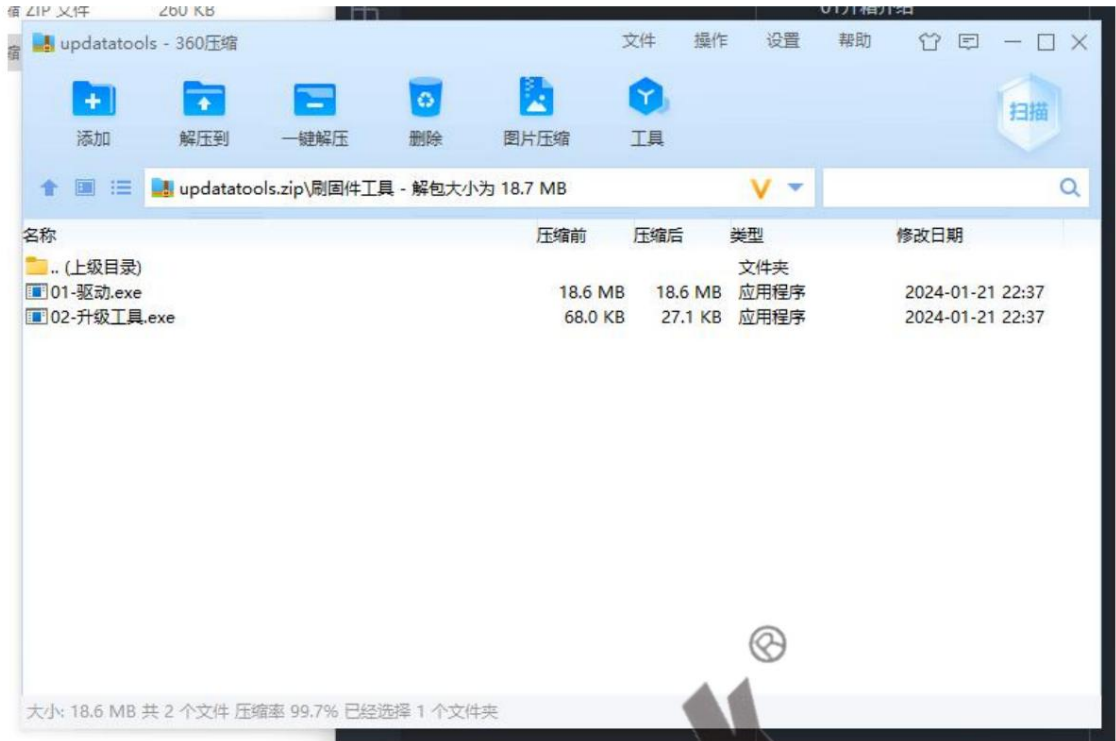
Upgrading is not mandatory; it's up to you whether you want to upgrade. If you're using it without any problems, you don't need to upgrade. Upgrading the firmware can fix bugs present in older firmware versions.

If you wish to upgrade, please read the following document carefully.

01_Download and upgrade drivers

The first time you upgrade the box, you need to install the upgrade driver on the computer used for the upgrade. Subsequent upgrades will not require this. Note that you need to install the driver the first time!

[Click here to download and upgrade firmware driver](#)



After downloading, extract the files and install the 01-driver. Exit the program once installation is complete.

02_Download the firmware that needs to be upgraded

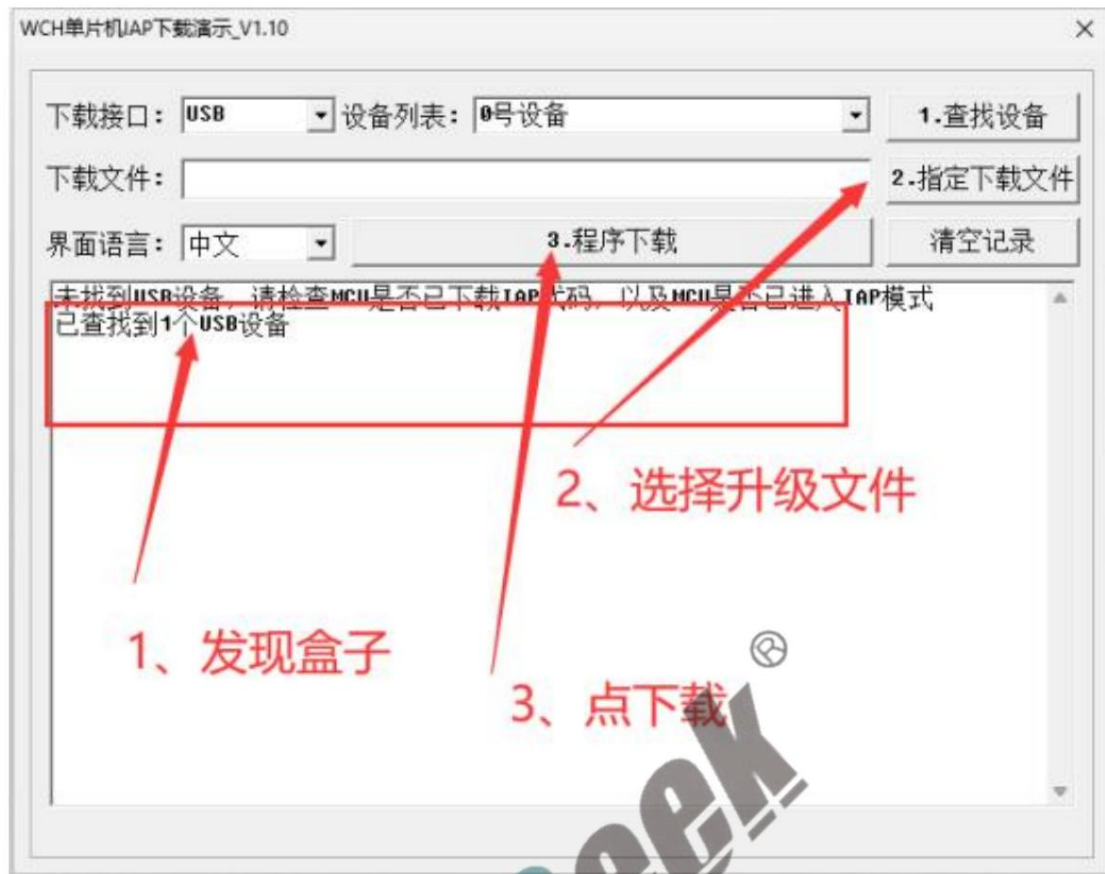
[Download the latest firmware here.](#)

[For other firmware versions, click here.](#)

03_Firmware Upgrade

Please be sure to follow the steps below strictly.

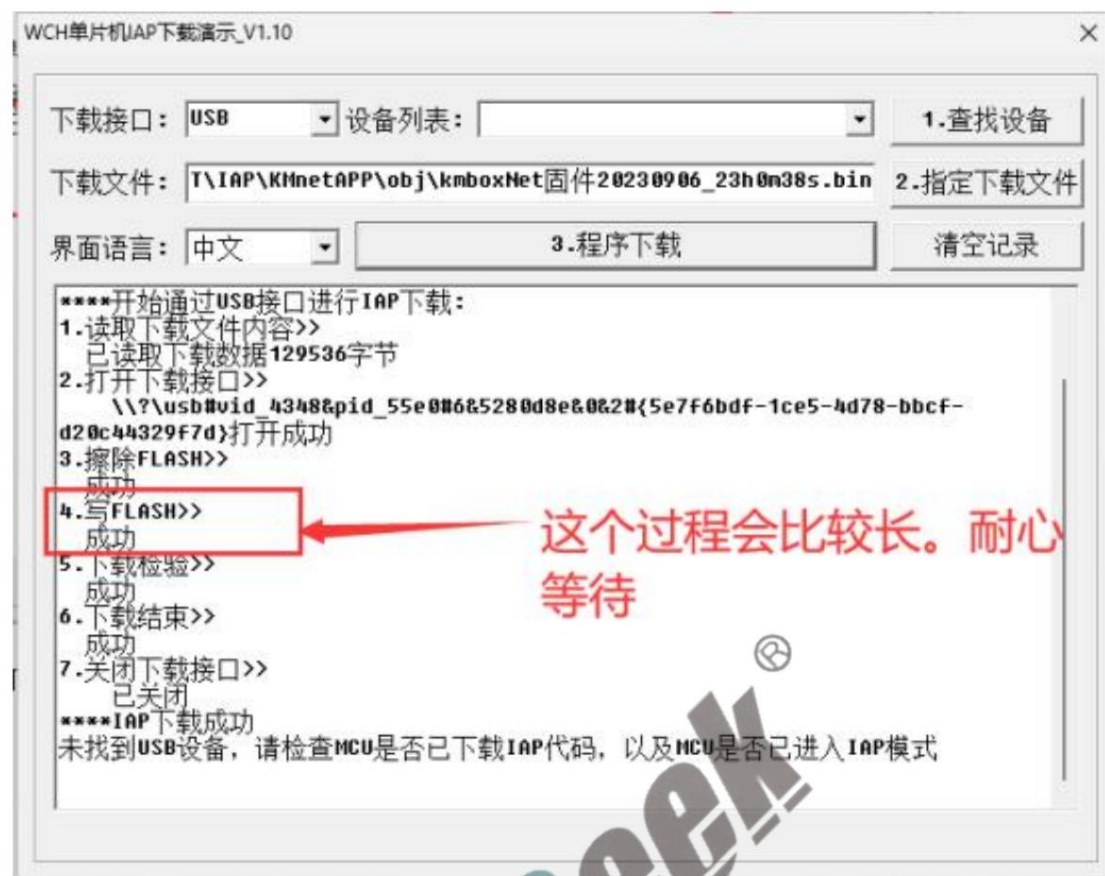
1. Disconnect all USB cables.
2. Press and hold the only button inside the box.
3. Connect the game console to the upgrade computer using a USB cable. You will see the network connection light flashing rapidly. At this point, you can release the button.
4. Open the upgrade tool WCHMcuIAP_WinAPP.exe (double-click to open).



Select the firmware you want to upgrade to. Note: If you cannot find the file when selecting the file, make sure to select "Bin" as the file type in the filter.

Click to download.

5. After successful download, the box will automatically restart, and the display screen will show normal operation.



If the upgrade fails, please repeat the above steps.

For a complete upgrade process, please refer to the video tutorial below:

Note: If you still have problems, please check if the main and secondary units are connected correctly. The game console port on the box should be connected to the computer on which you perform the upgrade. For example, if you downloaded the upgrade software on the secondary unit, connect the game console port to the secondary unit for the upgrade; if you downloaded it on the main unit, connect the game console port to the main unit for the upgrade.

How to modify a curve

01- What is the hardware curve correction function of the box?

The box has hardware curve correction disabled by default. This feature allows the mouse trajectory to avoid abrupt changes and achieve a smoother, more continuous trajectory.

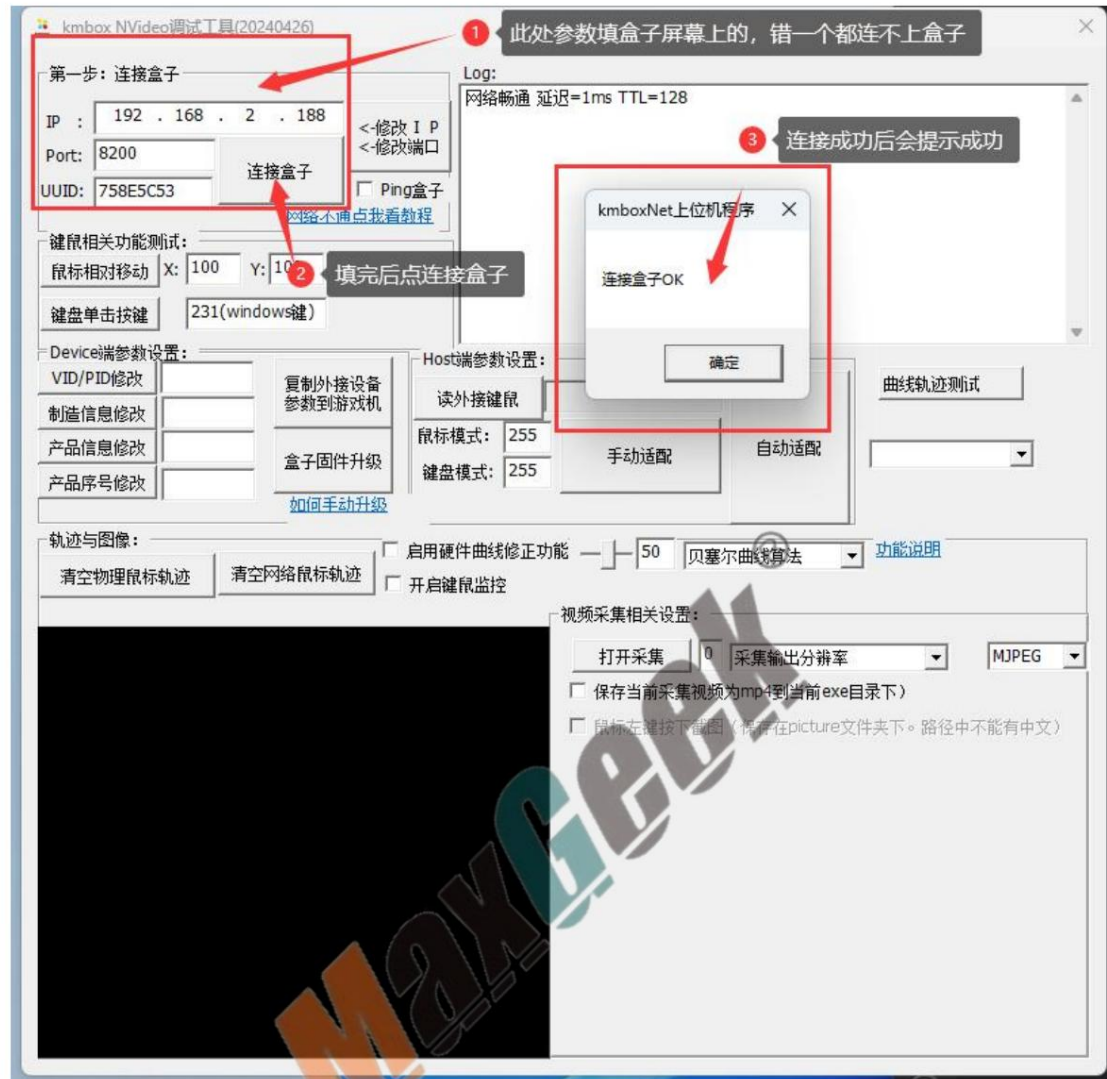
Note: Curves are off by default. To turn them off after enabling them, simply uncheck the box next to the curves feature.

The two videos below are explanations of curves.

02- Download the host computer control software for the box

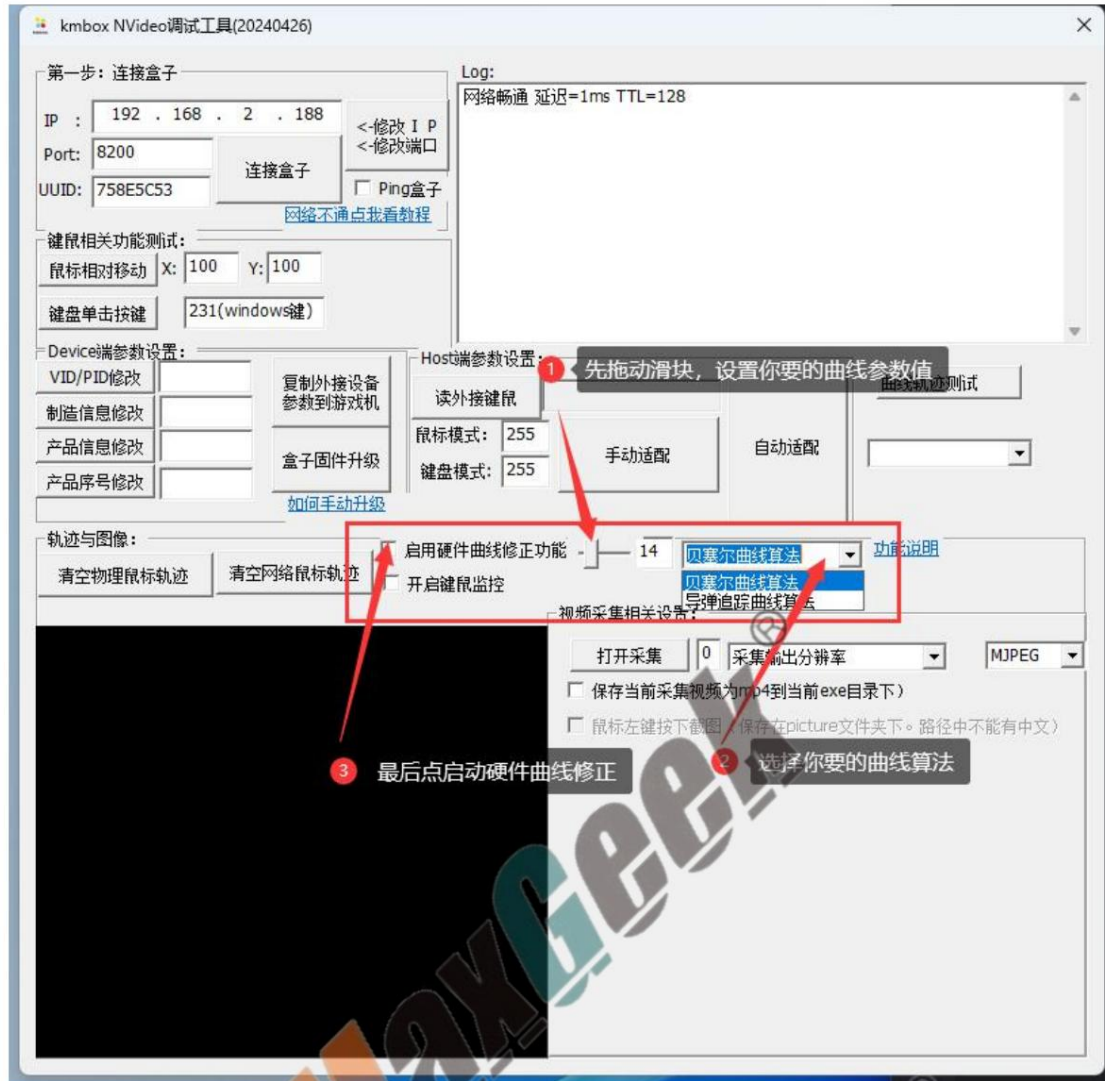
Modifying the curve first requires the box's host computer software. Click here to download [the kmboxNet host computer software](#).

03- Open the host computer software on the box



If you are unable to connect to the box, please refer to the beginner's tutorial and ensure that your network is working properly before using this software to connect to the box.

After a successful connection, you can select the hardware curve value and algorithm type (note that you must strictly follow steps 1, 2, and 3 below).



The box display looks like this after activation:



How to restore factory settings

If you accidentally mess up the TV box and the software can't connect to it, you can force the TV box to restore its factory settings.

The first method is to flash the firmware. Please refer to the firmware flashing tutorial.

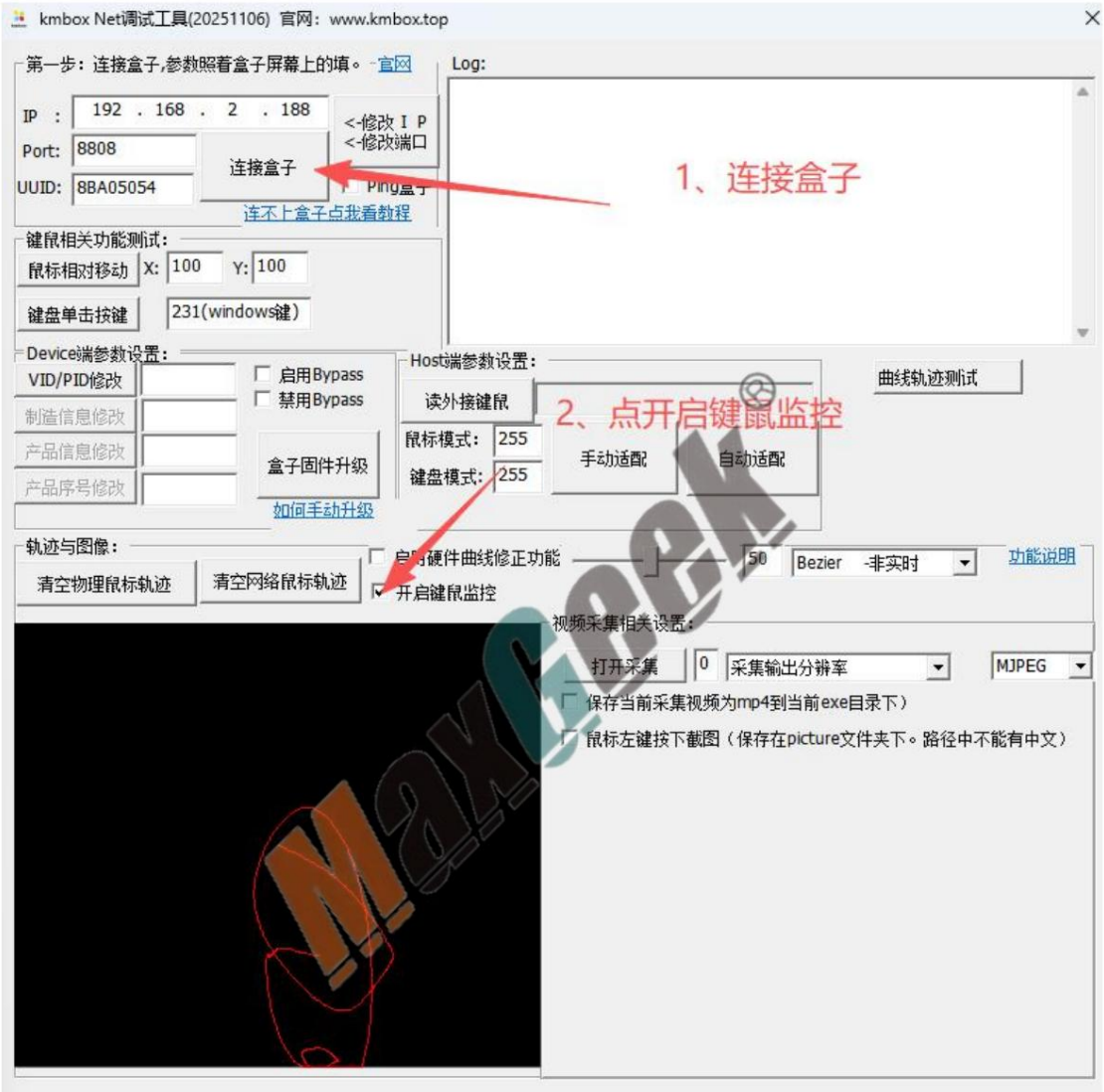
The second method is to press and hold the button inside the box until the screen flashes "recovery". This will restore the current version to factory settings.

You can use any of the methods above to restore factory settings.

How to enable the track display function

The trajectory display function is used to view the final movement trajectory of the box. It can help us analyze the root cause of abnormal keyboard and mouse data. It's generally used more by developers. Ordinary users can ignore it.

This feature can visually display mouse movement trajectories, including those of third-party software. To verify the validity of software movements, you can enable the monitoring function.



Note: If a system pop-up window asks whether to allow kmboxNet to use the network after enabling this feature, please click "Yes". Otherwise, the software will not store tracks.

Apply this knowledge to other situations. If there is no trace, first check if your system firewall is disabled, or if the executable file is allowed to access the network.

Frequently Asked Questions

1. Should I recommend Net or B+?

It depends on your personal needs. If you prefer to use API calls, I recommend .NET; if you prefer to write your own scripts, I recommend B+.

2. What is the difference between Net and B+?

The B+ version has onboard functionality (supports custom scripts). For other detailed differences, please refer to the respective introductions for the Net and B versions.

3. Problems encountered during the upgrade?

Check if the main and secondary units are connected correctly. Upgrade the software on the computer connected to the game console port of the set-top box. For example, if the upgrade software was downloaded on the secondary unit, connect it to the secondary unit for the upgrade; if it was downloaded on the main unit, connect it to the main unit for the upgrade.

4. Why is there still no response after installing the network card driver?

(1) Check if the network cable is connected. (2) Install the network card driver where the network USB cable is connected. Do not install the driver on computer A and connect the cable to computer B.

5. Why did I suddenly lose internet access after changing my IP address?

If you have no internet connection, it's likely that you've changed the network adapter settings incorrectly. Usually, a new network adapter (the box's network adapter) will appear on the page; you need to change the box's network adapter, not your own local network adapter. See the network adapter modification tutorial above for detailed instructions.

6. How do you close the curves of the box?

The curve function is off by default. To turn it off after enabling it, simply uncheck the box next to the curve function.

7. Why does the mouse sensitivity decrease when connected to the box?

You can adjust your mouse polling rate to below 1000 to match.

8. How do I change the mouse polling rate?

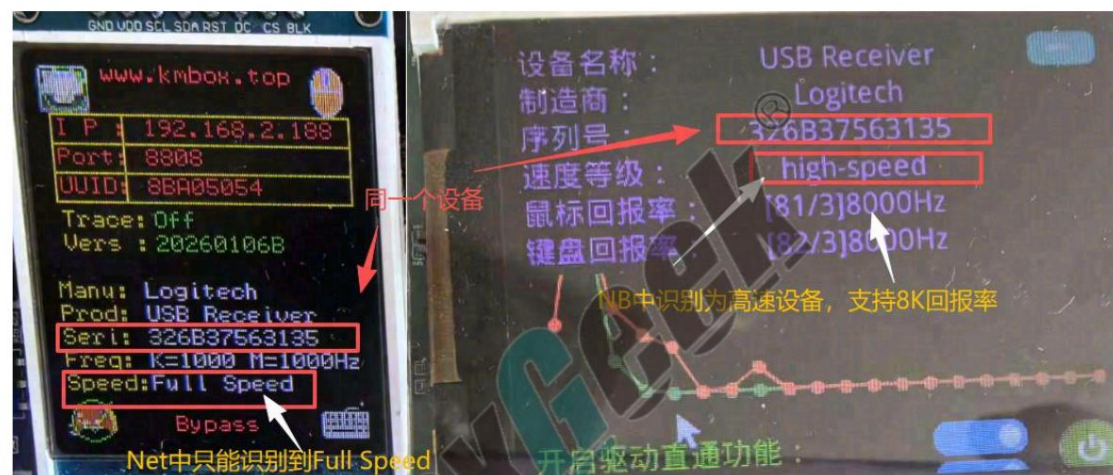
The values in the .NET version cannot be modified; you need to adjust your own mouse polling rate (refer to your mouse-related tutorials).

9. Why is the mouse noticeably unresponsive and has a large latency in bypass mode?

This situation usually occurs because the mouse is out of range. Net only supports Full Speed (12Mbps). If your device is High Speed (480Mbps), the Net box will only recognize it as Full Speed and parse it accordingly. The USB protocol specifies that Full Speed devices have a frame rate of 1ms, and High Speed devices have a frame rate of 0.125ms. If a high-speed device requires a 1kHz polling rate, its defined value is 8. However, the box can only recognize Full Speed and not High Speed. Therefore, the box reading 8 means reading data once every 8ms, resulting in a large latency issue.

Additionally, changing 8 to 1 violates the bypass pass-through rule and is not feasible—the essence of this problem is that the device exceeds the specifications supported by NET.

For perfect pass-through support, please select the NB version with a higher speed rating; if you want to use high-speed devices on the Net, please select the KM mode.



Software Developer Zone

To be updated